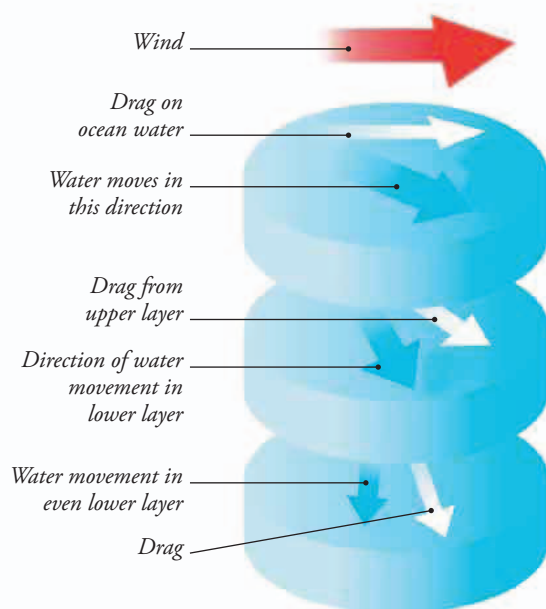


Surface currents

The wind that whips up waves also drives powerful surface currents. The main driving forces are the winds generated by global airflow and deflected by Earth's spin—the prevailing winds. The spin effect also influences the currents themselves, making them swerve right or left. As a result, they form huge rotating gyres that swirl around the oceans, carrying cold water into the tropics and warm water toward the poles.

WOW!

The Gulf Stream in the north Atlantic transports water at a rate of up to 5.3 billion cubic feet (150 million cubic meters) per second.



STRANGE EFFECTS

The spinning Earth effect that makes the wind swerve off-course does the same to ocean currents. They veer to the right in the northern hemisphere, and to the left in the southern hemisphere. Moving water at the surface drags deeper water with it, which swerves even further right or left. As a result, the current's direction changes with depth—a pattern called Ekman transport.

AROUND AND AROUND

The trade winds in the tropical Atlantic north of the equator blow toward the southwest, but Ekman transport pushes water westward. The current veers right as it comes up against North America, to become the Gulf Stream. This flows eastward, driven by prevailing winds blowing toward the northeast, then swerves south as the Canary Current. The resulting circulation is called the north Atlantic gyre. Similar gyres occur in all other oceans.

